GROWTHOS TELUGU

EPISODE 3 / 10

PROJECT J.A.R.V.I.S • BUILDING MY OWN AI ASSISTANT

# GIVING EARS & VOICE

### Offline Voice Assistant Setup Guide

| COST | INTERNET | DIFFICULTY | TIME |
|---|---|---|---|
| ₹0 | NONE | EASY | 30 MIN |

## WHAT WE BUILD IN THIS EPISODE

**FEATURES**

- Speech-to-Text (Ears) — Voice to text offline
- Audio Recording — Mic input reliably captured
- 100% Free & Open Source Stack

**BENEFITS**

- Text-to-Speech (Mouth) — JARVIS speaks back
- Zero API Keys — No subscriptions needed
- Works without Internet connection

## STEP 1 — INSTALL REQUIRED LIBRARIES

```
$ pip install faster-whisper pyttsx3 sounddevice numpy
```

Run this in VS Code Terminal

| Library | Purpose | Type |
|---|---|---|
| faster-whisper | Lightning-fast local speech recognition (STT) | AI Model |
| pyttsx3 | Offline text-to-speech — Windows built-in voices | TTS Engine |
| sounddevice | Microphone access & audio recording | Audio I/O |
| numpy | Audio data processing & normalization | Data Math |

**NOTE:** If you get errors, make sure Python is added to PATH (see Episode 2 PDF)
Run the command again if ModuleNotFoundError appears

## PERFORMANCE BENCHMARKS

| RECORDING TIME | TRANSCRIPTION TIME | TTS RESPONSE TIME |
|---|---|---|
| **3 seconds** | **~0.5 – 1 second (CPU)** | **~0.3 seconds** |
| **TOTAL LATENCY** | **RAM USAGE** | **INTERNET REQUIRED** |
| **~2 – 3 seconds** | **~200 MB** | **NONE** |

**voice_test.py — Complete Source Code**

```
import sounddevice as sd
import numpy as np
from faster_whisper import WhisperModel
import pyttsx3

print("  Listening... Speak now (3 seconds)!")

# 1. Record Audio
samplerate = 16000
duration = 3
print(f"  Recording for {duration} seconds...")

recording = sd.rec(int(duration * samplerate),
                   samplerate=samplerate,
                   channels=1,
                   dtype="int16")
sd.wait()
print("  Recording complete!")

# 2. Convert to float32 for Whisper (normalize)
audio_float = recording.flatten().astype(np.float32) / 32768.0

# 3. Transcribe Speech to Text
print("  Processing audio...")
model = WhisperModel("tiny", device="cpu", compute_type="int8")
segments, info = model.transcribe(audio_float, beam_size=1, language="en")

text = "".join([segment.text for segment in segments])
print(f"  You said: {text.strip()}")

# 4. Text to Speech Response
if text.strip():
    print("  Jarvis Speaking...")
    engine = pyttsx3.init()
    engine.setProperty("rate", 150)  # Speed of speech
    engine.say(text)
    engine.runAndWait()
else:
    print("  No speech detected. Try speaking louder!")

print("  Task Complete!")
```

## STEP 3 — HOW TO RUN

**01** Save the code as voice_test.py in your jarvis_Ai folder

**02** Open VS Code Terminal (Ctrl + ` )

**03** Type: python voice_test.py and press Enter

**04** Wait for the "Listening..." message to appear

**05** Speak clearly into your mic for 3 seconds

**06** Watch JARVIS transcribe AND speak back your words!

## TROUBLESHOOTING GUIDE

**PortAudioError**
Enable mic: Windows Privacy Settings → Microphone → "Let desktop apps access microphone"

**No speech detected**
Speak LOUDER & CLEARER. Check mic volume in Sound Settings (Windows)

**ModuleNotFoundError**
Run `pip install faster-whisper pyttsx3 sounddevice numpy` in terminal again

**Memory Error**
Reduce duration from 3 to 2 seconds in the recording section of code

**Robotic voice**
Install voices: Windows Settings → Time & Language → Speech → Add voices

## KEY CONCEPTS EXPLAINED

**Why dtype='int16'?**
Records audio in 16-bit format (half memory of float32). Converted to float32 before Whisper.

**Why divide by 32768.0?**
Normalizes int16 audio (-32768 to 32767) to float range (-1.0 to 1.0) that Whisper expects.

**Why beam_size=1?**
Faster transcription with minimal accuracy loss. Perfect for real-time voice commands.

**Why language="en"?**
Forces English detection. Remove for auto-detection (supports Telugu/Hindi but slower).

## SERIES ROADMAP — 10 EPISODES

| | | |
|---|---|---|
| EP 1 | Vision & Blueprint | DONE |
| EP 2 | Installing the Brain (Ollama + Python) | DONE |
| EP 3 | **Giving Ears & Voice** YOU ARE HERE | CURRENT |
| EP 4 | Wake Word Detection | UPCOMING |
| EP 5 | Full Conversation Loop | UPCOMING |
| EP 6 | Memory System (ChromaDB) | UPCOMING |
| EP 7 | PC Control & Automation | UPCOMING |
| EP 8 | Smart Automation Pipelines | UPCOMING |
| EP 9 | Futuristic JARVIS UI | UPCOMING |
| EP 10 | Grand Finale — Full JARVIS Live! | UPCOMING |

***"Building the future, one line of code at a time."***
— Growth OS Telugu | Project J.A.R.V.I.S